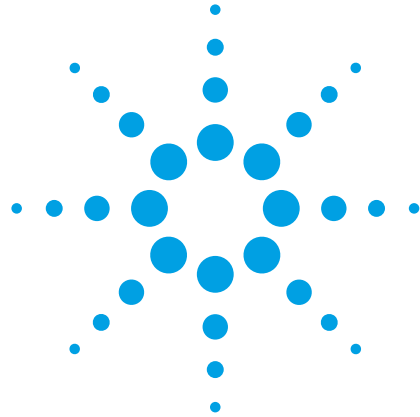


Agilent B1500A Semiconductor Device Analyzer

Data Sheet



Introduction

The Agilent B1500A Semiconductor Device Analyzer with EasyEXPERT software is a complete parametric test solution. It supports all aspects of parametric test, from basic manual measurement to test automation across a wafer in conjunction with a semiautomatic wafer prober. Because the B1500A utilizes the Microsoft® Windows® 7 operating system, it integrates easily into your PC-based work environment. Best of all, the familiar Windows graphical user interface (GUI) and convenient online help menus minimize the need for instrument training.



Agilent Technologies

Basic Features

- PC-based instrument with touch screen interface; optional USB keyboard and mouse available
- EasyEXPERT software with over 300 categorized application tests supplied with the instrument
- Performs current versus voltage (IV) measurements
- Performs capacitance versus voltage (CV), capacitance versus time (Ct) and capacitance versus frequency (Cf) measurements
- Performs quasi-static CV (QSCV) measurements
- High voltage pulse forcing up to 40 V with ALWG and voltage monitor capabilities.
- Performs fast IV measurement synchronized with the applied waveforms for accurate transient IV or time-domain measurements
- Front-panel Classic Test measurement modes supported: single-channel sweep, multi-channel sweep, time sampling, list sweep, CV sweep and direct control (GPIB FLEX)
- GUI-based control of the Agilent B2200A, and B2201A switching matrices
- Modular mainframe with ten module slots and one 4.2 A ground unit
- Multiple source/monitor unit (SMU) types available: medium power (MPSMU), high-power (HPSMU), and high-resolution (HRSMU)
- Multi-frequency capacitance measurement unit (MFCMU) available

- High-voltage semiconductor pulse generator unit (HV-SPGU) available
- Waveform generator/fast measurement unit (WGFMU) available
- High-resolution, analog-to-digital converter (ADC) available to all installed SMUs
- High-speed ADC present on each installed SMU
- SMU/AUX path switching supported on the atto-sense and switch unit (ASU)
- SMU/MFCMU switching supported using SMU CMU unify unit (SCUU) and guard switch unit (GSWU)
- MFCMU automatically identifies capacitance measurement accessories
- WGFMU/SMU path switching supported on the remote-sense and switch unit (RSU)
- GPIB port for instrument control
- Self-test, self-calibration, diagnostics

Specification conditions

This document lists specifications and supplemental information for the B1500A and its associated modules. The specifications are the standards against which the B1500A and its associated modules are tested. When the B1500A and any of its associated modules are shipped from the factory, they meet the specifications. The “supplemental” information and “typical” entries in the following specifications are not warranted, but provide useful information about the functions and performance of the instrument.

The measurement and output accuracy are specified at the rear panel connector terminals when referenced to the Zero Check terminal. The B1530A WGFMU measurement and output accuracy are specified at the output terminal of the RSU. Accuracy is specified under the following conditions:

1. Temperature: 23 °C ±5 °C
2. Humidity: 20 % to 60 %
3. After 40 minute warm-up followed by self-calibration
4. Ambient temperature change less than ±1 °C after self-calibration execution, not applicable for MFCMU and WGFMU
5. Measurement made within one hour after self-calibration execution, not applicable for MFCMU and WGFMU
6. Calibration period: 1 year
7. SMU integration time setting:
1 PLC (1 nA to 1A range, voltage range)
20 PLC (100 pA range)
50 PLC (1 pA to 10 pA range)
Averaging of high-speed ADC:
128 samples per 1 PLC
8. SMU filter: ON (for SMUs)
9. SMU measurement terminal connection: Kelvin connection
10. WGFMU load capacitance: 25 pF or less

Note: Agilent Technologies is responsible for removing, installing, and replacing the B1500A modules. Contact your nearest Agilent Technologies to install and calibrate the B1500A modules

B1500A specification

Supported plug-in modules

The B1500A supports ten slots for plug-in modules.

Part number	Description	Slots occupied	Range of operation	Measure resolution
B1510A	High power source/monitor unit (HPSMU)	2	-200 V to 200 V, -1 A to 1 A	2 µV, 10 fA
B1511A	Medium power source/monitor unit (MPSMU)	1	-100 V to 100 V, -100 mA to 100 mA	0.5 µV, 10 fA
B1517A	High resolution source/monitor unit (HRSMU)	1	-100 V to 100 V, -100 mA to 100 mA	0.5 µV, 1 fA
E5288A ¹	Atto-sense and switch unit (ASU)	—	-100 V to 100 V, -100 mA to 100 mA	0.5 µV, 100 aA
B1520A	Multi frequency capacitance measurement unit (MFCMU)	1	1 kHz to 5 MHz	0.035 fFrms ²
B1525A	High voltage semiconductor pulse generator unit (HV-SPGU)	1	±40 V (80 Vp-p)	50 µV
B1530A	Waveform generator/fast measurement unit (WGFMU)	1	PG Mode: -3 V to 3 V, -5 V to 5 V Fast IV Mode: -3 V to 3 V, -5 V to 5 V, 0 V to 10 V, -10 V to 0 V	0.014 % of the range ³

1. This is connected with the B1517A high resolution SMU.

2. Dispersion of measurement values when connecting a DUT 10 pF to the measurement terminals under the measurement condition of frequency 1 MHz, signal level 250 mVac, and measurement time 1 PLC. The display resolution is 0.000001 fF at 1 fF order by 6 digits display.

3. Display resolution

Maximum module configuration

The B1500A can contain up to 4 dual-slot SMUs (HPSMUs) and 2 single-slot SMUs (MPSMUs and/or HRSMUs); it can contain up to 10 single-slot SMUs (MPSMUs and/or HRSMUs); and it can contain any combination of dual-slot and single-slot SMUs between these two extremes. Only one single-slot MFCMU can be installed per B1500A mainframe. Up to five single-slot HV-SPGUs can be installed per mainframe. Up to five single-slot WGFMUs can be installed per mainframe.

When one or more WGFMU modules are installed in the B1500A mainframe, the following table applies. Multiply the values given below by the number of installed modules of that type and add the products together. The sum of the products must be less than or equal to 59 for the configuration to be permissible.

MPSMU	2
HRSMU	2
HPSMU	14
MFCMU	7
HV-SPGU	12
WGFMU	10

Maximum voltage between common and ground

≤ ±42 V

Ground unit (GNDU) specification

The GNDU is furnished standard with the B1500A mainframe.

Output voltage: 0 V ±100 μV

Maximum sink current: ±4.2 A

Output terminal/connection:

Triaxial connector, Kelvin (remote sensing)

GNDU supplemental information

Load capacitance: 1 μF

Cable resistance:

For $I_s \leq 1.6$ A: force line $R < 1 \Omega$

For $1.6 \text{ A} < I_s \leq 2.0$ A: force line $R < 0.7 \Omega$

For $2.0 \text{ A} < I_s \leq 4.2$ A: force line $R < 0.35 \Omega$

For all cases: sense line $R \leq 10 \Omega$

Where I_s is the current being sunk by the GNDU.

MPSMU and HRSMU module specifications

Voltage range, resolution, and accuracy (high resolution ADC)

Voltage range	Force resolution	Measure resolution	Force accuracy ¹	Measure accuracy ¹	Maximum current
±0.5 V	25 μV	0.5 μV	±(0.018 % + 150 μV)	±(0.01 % + 120 μV)	100 mA
±2 V	100 μV	2 μV	±(0.018 % + 400 μV)	±(0.01 % + 140 μV)	100 mA
±5 V	250 μV	5 μV	±(0.018 % + 750 μV)	±(0.009 % + 250 μV)	100 mA
±20 V	1 mV	20 μV	±(0.018 % + 3 mV)	±(0.009 % + 900 μV)	100 mA
±40 V	2 mV	40 μV	±(0.018 % + 6 mV)	±(0.01 % + 1 mV)	²
±100 V	5 mV	100 μV	±(0.018 % + 15 mV)	±(0.012 % + 2.5 mV)	²

1. ± (% of read value + offset voltage V)

2. 100 mA ($V_o \leq 20$ V), 50 mA ($20 \text{ V} < V_o \leq 40$ V), 20 mA ($40 \text{ V} < V_o \leq 100$ V), V_o is the output voltage in Volts.

Current range, resolution, and accuracy (high resolution ADC)

SMU type	Current range	Force resolution	Measure resolution ^{1,2}	Force accuracy ³	Measure accuracy ³	Maximum voltage
HRSMU w/ ASU	±1 pA	1 fA	100 aA	±(0.9 % + 15 fA)	±(0.9 % + 12 fA)	100 V
	±10 pA	5 fA	400 aA (with ASU) 1 fA (HRSMU)	±(0.46 % + 30 fA + 10 aA x V_o)	±(0.46 % + 15 fA + 10 aA x V_o)	100 V
	±100 pA	5 fA	500 aA (with ASU) 2 fA (HRSMU)	±(0.3 % + 100 fA + 100 aA x V_o)	±(0.3 % + 30 fA + 100 aA x V_o)	100 V
	±1 nA	50 fA	10 fA	±(0.1 % + 300 fA + 1 fA x V_o)	±(0.1 % + 200 fA + 1 fA x V_o)	100 V
	±10 nA	500 fA	10 fA	±(0.1 % + 3 pA + 10 fA x V_o)	±(0.1 % + 1 pA + 10 fA x V_o)	100 V
	±100 nA	5 pA	100 fA	±(0.05 % + 30 pA + 100 fA x V_o)	±(0.05 % + 20 pA + 100 fA x V_o)	100 V
	±1 μA	50 pA	1 pA	±(0.05 % + 300 pA + 1 pA x V_o)	±(0.05 % + 100 pA + 1 pA x V_o)	100 V
	±10 μA	500 pA	10 pA	±(0.05 % + 3 nA + 10 pA x V_o)	±(0.04 % + 2 nA + 10 pA x V_o)	100 V
	±100 μA	5 nA	100 pA	±(0.035 % + 15 nA + 100 pA x V_o)	±(0.03 % + 3 nA + 100 pA x V_o)	100 V
	±1 mA	50 nA	1 nA	±(0.04 % + 150 nA + 1 nA x V_o)	±(0.03 % + 60 nA + 1 nA x V_o)	100 V
MPSMU	±10 mA	500 nA	10 nA	±(0.04 % + 1.5 μA + 10 nA x V_o)	±(0.03 % + 200 nA + 10 nA x V_o)	100 V
	±100 mA	5 μA	100 nA	±(0.045 % + 15 μA + 100 nA x V_o)	±(0.04 % + 6 μA + 100 nA x V_o)	⁴

1. Specified measurement resolution is limited by fundamental noise limits. Minimum displayed resolution is 1 aA at 1 pA range by 6 digits.

2. Measurements made in the lower ranges can be greatly impacted by vibrations and shocks. These specifications assume an environment free of these factors.

3. ± (% of read value + offset current (fixed part determined by the output/measurement range + proportional part that is multiplied by V_o))

4. 100 V ($I_o \leq 20$ mA), 40 V ($20 \text{ mA} < I_o \leq 50$ mA), 20 V ($50 \text{ mA} < I_o \leq 100$ mA), I_o is the output current in Amps.

Voltage range, resolution, and accuracy (high speed ADC)

Voltage range	Force resolution	Measure resolution	Force accuracy ¹	Measure accuracy ¹	Maximum current
±0.5 V	25 µV	25 µV	±(0.018 % + 150 µV)	±(0.01 % + 250 µV)	100 mA
±2 V	100 µV	100 µV	±(0.018 % + 400 µV)	±(0.01 % + 700 µV)	100 mA
±5 V	250 µV	250 µV	±(0.018 % + 750 µV)	±(0.01 % + 2 mV)	100 mA
±20 V	1 mV	1 mV	±(0.018 % + 3 mV)	±(0.01 % + 4 mV)	100 mA
±40 V	2 mV	2 mV	±(0.018 % + 6 mV)	±(0.015 % + 8 mV)	²
±100 V	5 mV	5 mV	±(0.018 % + 15 mV)	±(0.02 % + 20 mV)	²

1. ± (% of read value + offset voltage V)

2. 100 mA ($V_o \leq 20$ V), 50 mA (20 V $< V_o \leq 40$ V), 20 mA (40 V $< V_o \leq 100$ V), V_o is the output voltage in Volts.

Current range, resolution, and accuracy (high speed ADC)

SMU type	Current range	Force resolution	Measure resolution ^{1,2}	Force accuracy ³	Measure accuracy ³	Maximum voltage
HRSMU w/ ASU	±1 pA	1 fA	100 aA	±(0.9 % + 15 fA)	±(1.8 % + 12 fA)	100 V
HRSMU	±10 pA	5 fA	1 fA	±(0.46 % + 30 fA + 10 aA x V_o)	±(0.5 % + 15 fA + 10 aA x V_o)	100 V
	±100 pA	5 fA	5 fA	±(0.3 % + 100 fA + 100 aA x V_o)	±(0.5 % + 40 fA + 100 aA x V_o)	100 V
	MPSMU	±1 nA	50 fA	±(0.1 % + 300 fA + 1 fA x V_o)	±(0.25 % + 300 fA + 1 fA x V_o)	100 V
		±10 nA	500 fA	±(0.1 % + 3 pA + 10 fA x V_o)	±(0.25 % + 2 pA + 10 fA x V_o)	100 V
		±100 nA	5 pA	±(0.05 % + 30 pA + 100 fA x V_o)	±(0.1 % + 20 pA + 100 fA x V_o)	100 V
		±1 µA	50 pA	±(0.05 % + 300 pA + 1 pA x V_o)	±(0.1 % + 200 pA + 1 pA x V_o)	100 V
		±10 µA	500 pA	±(0.05 % + 3 nA + 10 pA x V_o)	±(0.05 % + 2 nA + 10 pA x V_o)	100 V
		±100 µA	5 nA	±(0.035 % + 15 nA + 100 pA x V_o)	±(0.05 % + 20 nA + 100 pA x V_o)	100 V
		±1 mA	50 nA	±(0.04 % + 150 nA + 1 nA x V_o)	±(0.04 % + 200 nA + 1 nA x V_o)	100 V
		±10 mA	500 nA	±(0.04 % + 1.5 µA + 10 nA x V_o)	±(0.04 % + 2 µA + 10 nA x V_o)	100 V
		±100 mA	5 µA	±(0.045 % + 15 µA + 100 nA x V_o)	±(0.1 % + 20 µA + 100 nA x V_o)	⁴

1. Specified measurement resolution is limited by fundamental noise limits. Minimum displayed resolution is 1 aA at 1 pA range by 6 digits.

2. Measurements made in the lower ranges can be greatly impacted by vibrations and shocks. These specifications assume an environment free of these factors.

3. ± (% of read value + offset current (fixed part determined by the output/measurement range + proportional part that is multiplied by V_o))

4. 100 V ($I_o \leq 20$ mA), 40 V (20 mA $< I_o \leq 50$ mA), 20 V (50 mA $< I_o \leq 100$ mA), I_o is the output current in Amps.

Power consumption

Voltage source mode

Voltage range	Power
0.5 V	20 x I_c (W)
2 V	20 x I_c (W)
5 V	20 x I_c (W)
20 V	20 x I_c (W)
40 V	40 x I_c (W)
100 V	100 x I_c (W)

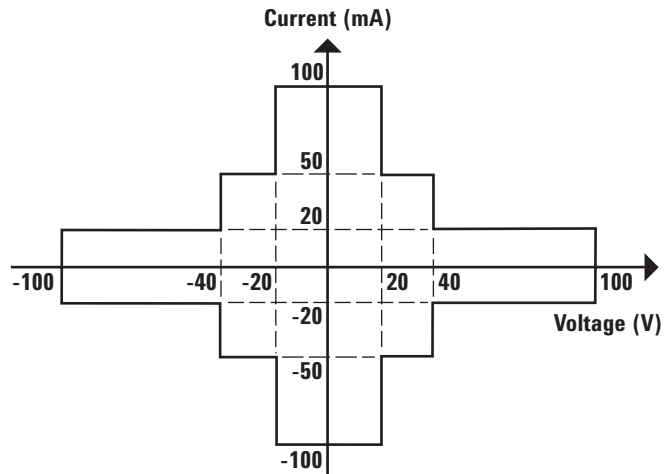
Where I_c is the current compliance setting.

Current source mode

Voltage compliance	Power
$V_c \leq 20$	20 x I_o (W)
$20 < V_c \leq 40$	40 x I_o (W)
$40 < V_c \leq 100$	100 x I_o (W)

Where V_c is the voltage compliance setting and I_o is output current.

MPSMU and HRSMU measurement and output range



HPSMU module specifications

Voltage range, resolution, and accuracy (high resolution ADC)

Voltage range	Force resolution	Measure resolution	Force accuracy ¹	Measure accuracy ¹	Maximum current
±2 V	100 µV	2 µV	±(0.018 % + 400 µV)	±(0.01 % + 140 µV)	1 A
±20 V	1 mV	20 µV	±(0.018 % + 3 mV)	±(0.01 % + 140 µV)	1 A
±40 V	2 mV	40 µV	±(0.018 % + 6 mV)	±(0.01 % + 1 mV)	500 mA
±100 V	5 mV	100 µV	±(0.018 % + 15 mV)	±(0.012 % + 2.5 mV)	125 mA
±200 V	10 mV	200 µV	±(0.018 % + 30 mV)	±(0.014 % + 2.8 mV)	50 mA

1. ± (% of read value + offset voltage V)

Current range, resolution, and accuracy (high resolution ADC)

Current range	Force resolution	Measure resolution ¹	Force accuracy ²	Measure accuracy ²	Maximum voltage
±1 nA	50 fA	10 fA	±(0.1 %+300 fA+1 fA x Vo)	±(0.1 %+300 fA+1 fA x Vo)	200 V
±10 nA	500 fA	10 fA	±(0.1 %+3 pA+10 fA x Vo)	±(0.1 %+2.5 pA+10 fA x Vo)	200 V
±100 nA	5 pA	100 fA	±(0.05 %+30 pA+100 fA x Vo)	±(0.05 %+25 pA+100 fA x Vo)	200 V
±1 µA	50 pA	1 pA	±(0.05 %+300 pA+1 pA x Vo)	±(0.05 %+100 pA+1 pA x Vo)	200 V
±10 µA	500 pA	10 pA	±(0.05 %+3 nA+10 pA x Vo)	±(0.04 %+2 nA+10 pA x Vo)	200 V
±100 µA	5 nA	100 pA	±(0.035 %+15 nA+100 pA x Vo)	±(0.03 %+3 nA+100 pA x Vo)	200 V
±1 mA	50 nA	1 nA	±(0.04 %+150 nA+1 nA x Vo)	±(0.03 %+60 nA+1 nA x Vo)	200 V
±10 mA	500 nA	10 nA	±(0.04 %+1.5 nA+10 nA x Vo)	±(0.03 %+200 nA+10 nA x Vo)	200 V
±100 mA	5 µA	100 nA	±(0.045 %+15 µA+100 nA x Vo)	±(0.04 %+6 µA+100 nA x Vo)	³
±1 A	50 µA	1 µA	±(0.4 %+300 µA+1 µA x Vo)	±(0.4 %+150 µA+1 µA x Vo)	³

1. Specified measurement resolution is limited by fundamental noise limits.

2. ± (% of read value + offset current (fixed part determined by the output/measurement range + proportional part that is multiplied by Vo))

3. 200 V (Io ≤ 50 mA), 100 V (50 mA < Io ≤ 125 mA), 40 V (125 mA < Io ≤ 500 mA), 20 V (500 mA < Io ≤ 1 A), Io is the output current in Amps.

Voltage range, resolution, and accuracy (high speed ADC)

Voltage range	Force resolution	Measure resolution	Force accuracy ¹	Measure accuracy ¹	Maximum current
±2 V	100 µV	2 µV	±(0.018 % + 400 µV)	±(0.01 % + 700 µV)	1 A
±20 V	1 mV	20 µV	±(0.018 % + 3 mV)	±(0.01 % + 4 mV)	1 A
±40 V	2 mV	40 µV	±(0.018 % + 6 mV)	±(0.015 % + 8 mV)	500 mA
±100 V	5 mV	100 µV	±(0.018 % + 15 mV)	±(0.02 % + 20 mV)	125 mA
±200 V	10 mV	200 µV	±(0.018 % + 30 mV)	±(0.035 % + 40 mV)	50 mA

1. ± (% of read value + offset voltage V)

Current range, resolution, and accuracy (high speed ADC)

Current range	Force resolution	Measure resolution ¹	Force accuracy ²	Measure accuracy ²	Maximum voltage
±1 nA	50 fA	10 fA	$\pm(0.1\% + 300 \text{ fA} + 1 \text{ fA} \times V_o)$	$\pm(0.25\% + 300 \text{ fA} + 1 \text{ fA} \times V_o)$	200 V
±10 nA	500 fA	10 fA	$\pm(0.1\% + 3 \text{ pA} + 10 \text{ fA} \times V_o)$	$\pm(0.25\% + 2 \text{ pA} + 10 \text{ fA} \times V_o)$	200 V
±100 nA	5 pA	100 fA	$\pm(0.05\% + 30 \text{ pA} + 100 \text{ fA} \times V_o)$	$\pm(0.1\% + 20 \text{ pA} + 100 \text{ fA} \times V_o)$	200 V
±1 µA	50 pA	1 pA	$\pm(0.05\% + 300 \text{ pA} + 1 \text{ pA} \times V_o)$	$\pm(0.1\% + 200 \text{ pA} + 1 \text{ pA} \times V_o)$	200 V
±10 µA	500 pA	10 pA	$\pm(0.05\% + 3 \text{ nA} + 10 \text{ pA} \times V_o)$	$\pm(0.05\% + 2 \text{ nA} + 10 \text{ pA} \times V_o)$	200 V
±100 µA	5 nA	100 pA	$\pm(0.035\% + 15 \text{ nA} + 100 \text{ pA} \times V_o)$	$\pm(0.05\% + 20 \text{ nA} + 100 \text{ pA} \times V_o)$	200 V
±1 mA	50 nA	1 nA	$\pm(0.04\% + 150 \text{ nA} + 1 \text{ nA} \times V_o)$	$\pm(0.04\% + 200 \text{ nA} + 1 \text{ nA} \times V_o)$	200 V
±10 mA	500 nA	10 nA	$\pm(0.04\% + 1.5 \text{ µA} + 10 \text{ nA} \times V_o)$	$\pm(0.04\% + 2 \text{ µA} + 10 \text{ nA} \times V_o)$	200 V
±100 mA	5 µA	100 nA	$\pm(0.045\% + 15 \text{ µA} + 100 \text{ nA} \times V_o)$	$\pm(0.1\% + 20 \text{ µA} + 100 \text{ nA} \times V_o)$	³
±1 A	50 µA	1 µA	$\pm(0.4\% + 300 \text{ µA} + 1 \text{ µA} \times V_o)$	$\pm(0.5\% + 300 \text{ µA} + 1 \text{ µA} \times V_o)$	³

1. Specified measurement resolution is limited by fundamental noise limits.

2. $\pm$ (% of read value + offset current (fixed part determined by the output/measurement range + proportional part that is multiplied by V_o))

3. 200 V ($I_o \leq 50 \text{ mA}$), 100 V ($50 \text{ mA} < I_o \leq 125 \text{ mA}$), 40 V ($125 \text{ mA} < I_o \leq 500 \text{ mA}$), 20 V ($500 \text{ mA} < I_o \leq 1 \text{ A}$), I_o is the output current in Amps.

Power consumption

Voltage source mode

Voltage range	Power
2 V	$20 \times I_c \text{ (W)}$
20 V	$20 \times I_c \text{ (W)}$
40 V	$40 \times I_c \text{ (W)}$
100 V	$100 \times I_c \text{ (W)}$
200 V	$200 \times I_c \text{ (W)}$

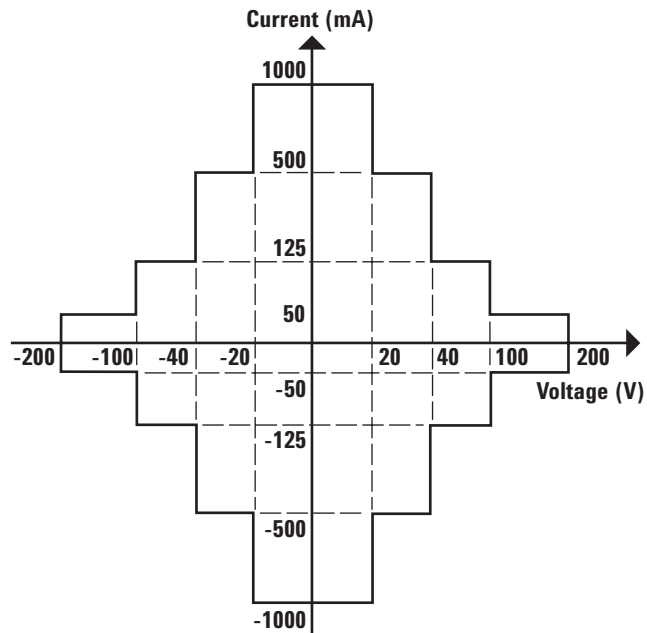
Where I_c is the current compliance setting.

Current source mode

Voltage compliance	Power
$V_c \leq 20$	$20 \times I_o \text{ (W)}$
$20 < V_c \leq 40$	$40 \times I_o \text{ (W)}$
$40 < V_c \leq 100$	$100 \times I_o \text{ (W)}$
$100 < V_c \leq 200$	$200 \times I_o \text{ (W)}$

Where V_c is the voltage compliance setting and I_o is output current.

HPSMU measurement and output range



Output terminal/connection

Dual triaxial connector, Kelvin (remote sensing)

Voltage/current compliance (limiting)

The SMU can limit output voltage or current to prevent damaging the device under test.

Voltage:

0 V to ± 100 V (MPSMU, HRSMU)
0 V to ± 200 V (HPSMU)

Current:

± 10 fA to ± 100 mA (HRSMU with ASU)
 ± 100 fA to ± 100 mA (HRSMU)
 ± 1 pA to ± 100 mA (MPSMU)
 ± 1 pA to ± 1 A (HPSMU)

Compliance accuracy:

Same as the current or voltage set accuracy.

About measurement accuracy

RF electromagnetic field and SMU measurement accuracy:

SMU voltage and current measurement accuracy can be affected by RF electromagnetic field strengths greater than 3 V/m in the frequency range of 80 MHz to 1 GHz. The extent of this effect depends upon how the instrument is positioned and shielded.

Induced RF field noise and SMU measurement accuracy:

SMU voltage and current measurement accuracy can be affected by induced RF field noise strengths greater than 3 V_{rms} in the frequency range of 150 kHz to 80 MHz. The extent of this effect depends upon how the instrument is positioned and shielded.

Pulse measurement

Pulse width: 500 μ sec to 2 s

Pulse period: 5 ms to 5 s

Period $\geq$ width + 2 ms
(when width $\leq$ 100 ms)

Period $\geq$ width + 10 ms
(when width > 100 ms)

Pulse resolution: 100 μ s

SMU pulse setting accuracy
(fixed measurement range,
supplemental information):

Width: 0.5 % + 50 μ s
Period: 0.5 % + 100 μ s

Supplemental information

Maximum allowable cable resistance
(Kelvin connection):

Sense: 10 Ω

Force: 10 Ω ($\leq$ 100 mA), 1.5 Ω (>100 mA)

Voltage source output resistance:
(Force line, Non-Kelvin connection)
0.2 Ω (HPSMU)
0.3 Ω (MPSMU, HRSMU)

Voltage measurement input resistance:
 $\geq 10^{13}$ Ω

Current source output resistance:
 $\geq 10^{13}$ Ω (1 nA range)

Current compliance setting accuracy
(for opposite polarity):
For 1 pA to 10 nA ranges:
I setting accuracy ± 12 % of range

For 100 nA to 1 A ranges:
I setting accuracy ± 2.5 % of range

Maximum capacitive load:
1 pA to 10 nA ranges: 1000 pF
100 nA to 10 mA ranges: 10 nF
100 mA and 1 A ranges: 100 μ F

Maximum guard capacitance:
900 pF (HPSMU, MPSMU, HRSMU)
660 pF (HRSMU with ASU)

Maximum shield capacitance:
5000 pF (HPSMU, MPSMU, HRSMU)
3500 pF (HRSMU with ASU)

Maximum guard offset voltage:
 ± 1 mV (HPSMU)
 ± 3 mV (MPSMU, HRSMU)
 ± 4.2 mV (HRSMU with ASU, I_{out} $\leq$ 100 μ A)

Noise characteristics (filter ON):
Voltage source: 0.01 % of V range (rms.)
Current source: 0.1 % of I range (rms.)

Overshoot (typical, filter ON):
Voltage source: 0.03 % of V range
Current source: 1 % of I range

Range switching transient noise (filter ON):
Voltage ranging: 250 mV
Current ranging: 70 mV

Slew rate: 0.2 V/ μ s (maximum)

Voltage range, resolution, and accuracy (high speed ADC)

Voltage range	Measure resolution	Measure accuracy ^{1, 2}
± 0.5 V ³	25 μ V	$\pm(0.01\% + 250 \mu\text{V})$
± 2 V	100 μ V	$\pm(0.01\% + 700 \mu\text{V})$
± 5 V ³	250 μ V	$\pm(0.01\% + 2 \text{ mV})$
± 20 V	1 mV	$\pm(0.01\% + 4 \text{ mV})$
± 40 V	2 mV	$\pm(0.015\% + 8 \text{ mV})$
± 100 V	5 mV	$\pm(0.02\% + 20 \text{ mV})$
± 200 V ⁴	10 mV	$\pm(0.035\% + 40 \text{ mV})$

1. $\pm$ (% of read value + offset voltage V)

2. Averaging is 128 samples in 1 PLC.

3. Only for MPSMU and HRSMU.

4. Only for HPSMU.

Current range, resolution, and accuracy (high speed ADC)

Current range	Measure resolution ^{1, 2}	Measure accuracy ³
± 1 pA ⁴	100 aA	$\pm(1.8\% + 12 \text{ fA})$
± 10 pA ⁵	1 fA	$\pm(0.5\% + 15 \text{ fA} + 10 \text{ aA} \times V_o)$
± 100 pA ⁵	5 fA	$\pm(0.3\% + 30 \text{ fA} + 100 \text{ aA} \times V_o)$
± 1 nA	50 fA	$\pm(0.1\% + 300 \text{ fA} + 1 \text{ fA} \times V_o)$
± 10 nA	500 fA	$\pm(0.1\% + 2 \text{ pA} + 10 \text{ fA} \times V_o)$
± 100 nA	5 pA	$\pm(0.05\% + 20 \text{ pA} + 100 \text{ fA} \times V_o)$
± 1 μ A	50 pA	$\pm(0.05\% + 200 \text{ pA} + 1 \text{ pA} \times V_o)$
± 10 μ A	500 pA	$\pm(0.04\% + 2 \text{ nA} + 10 \text{ pA} \times V_o)$
± 100 μ A	5 nA	$\pm(0.03\% + 20 \text{ nA} + 100 \text{ pA} \times V_o)$
± 1 mA	50 nA	$\pm(0.03\% + 200 \text{ nA} + 1 \text{ nA} \times V_o)$
± 10 mA	500 nA	$\pm(0.03\% + 2 \mu\text{A} + 10 \text{ nA} \times V_o)$
± 100 mA	5 μ A	$\pm(0.04\% + 20 \mu\text{A} + 100 \text{ nA} \times V_o)$
± 1 A ⁶	50 μ A	$\pm(0.4\% + 300 \mu\text{A} + 1 \mu\text{A} \times V_o)$

1. Specified measurement resolution is limited by fundamental noise limits. Minimum displayed resolution is 1 aA at 1 pA range by 6 digits.

2. Measurements made in the lower ranges can be greatly impacted by vibrations and shocks. These specifications assume an environment free of these factors.

3. $\pm$ (% of read value + offset current (fixed part determined by the output/measurement range + proportional part that is multiplied by V_o))

4. 1 pA range is for HRSMU with ASU.

5. 10 pA range and 100 pA range is for HRSMU with or without ASU.

6. Only for HPSMU.

MFCMU (multi frequency capacitance measurement unit) module specifications

Measurement functions

Measurement parameters:

Cp-G, Cp-D, Cp-Q, Cp-Rp, Cs-Rs, Cs-D,
Cs-Q, Lp-G, Lp-D, Lp-Q, Lp-Rp, Ls-Rs,
Ls-D, Ls-Q, R-X, G-B, Z-θ, Y-θ

Ranging:

Auto and fixed

Measurement terminal:

Four-terminal pair configuration, four
BNC (female) connectors

Cable length:

1.5 m or 3 m, automatic identification of
accessories

Test signal

Frequency:

Range: 1 kHz to 5 MHz
Resolution: 1 mHz (minimum)
Accuracy: $\pm 0.008\%$

Output signal level:

Range: 10 mV_{rms} to 250 mV_{rms}

Resolution: 1 mV_{rms}

Accuracy:

$\pm(10.0\% + 1 \text{ mV}_{\text{rms}})$
at the measurement port of the MFCMU

$\pm(15.0\% + 1 \text{ mV}_{\text{rms}})$
at the measurement port of the MFCMU
cable (1.5 m or 3.0 m)

Output impedance: 50 Ω, typical

Signal level monitor:

Range: 10 mV_{rms} to 250 mV_{rms}

Accuracy (open load):

$\pm(10.0\% \text{ of reading} + 1 \text{ mV}_{\text{rms}})$
at the measurement port of the MFCMU

$\pm(15.0\% \text{ of reading} + 1 \text{ mV}_{\text{rms}})$
at the measurement port of the MFCMU
cable (1.5 m or 3 m)

DC bias function

DC bias:

Range: 0 to $\pm 25 \text{ V}$

Resolution: 1 mV

Accuracy: $\pm(0.5\% + 5.0 \text{ mV})$ at the
measurement port of the MFCMU or the
MFCMU cable (1.5 m or 3.0 m)

Maximum DC bias current (supplemental information)

Impedance range	Maximum DC bias current
50 Ω	10 mA
100 Ω	10 mA
300 Ω	10 mA
1 kΩ	1 mA
3 kΩ	1 mA
10 kΩ	100 μA
30 kΩ	100 μA
100 kΩ	10 μA
300 kΩ	10 μA

Output impedance: 50 Ω, typical

DC bias monitor:

Range: 0 to $\pm 25 \text{ V}$

Accuracy (open load):

$\pm(0.2\% \text{ of reading} + 10.0 \text{ mV})$ at the
measurement port of the MFCMU or the
MFCMU cable (1.5 m or 3.0 m)

Sweep characteristics

Available sweep parameters:

Oscillator level, DC bias voltage,
frequency

Sweep type: linear, log

Sweep mode: single, double

Sweep direction: up, down

Number of measurement points:

Maximum 1001 points

Measurement accuracy

The following parameters are used to
express the impedance measurement
accuracy at the measurement port of
the MFCMU or the MFCMU cable
(1.5 m or 3.0 m).

Z_x : Impedance measurement value (Ω)

D_x : Measurement value of D

$E = E_p' + (Z_s' / |Z_x| + Y_o' |Z_x|) \times 100 (\%)$

$E_p' = E_{PL} + E_{POSC} + E_p (\%)$

$Y_o' = Y_{OL} + Y_{OSC} + Y_o (S)$

$Z_s' = Z_{SL} + Z_{OSC} + Z_s (\Omega)$

|Z| accuracy

$\pm E (\%)$

θ accuracy

$\pm E/100 (\text{rad})$

C accuracy

at $D_x \leq 0.1$

$\pm E (\%)$

at $D_x > 0.1$

$\pm E \times \sqrt{(1 + D_x^2)} (\%)$

D accuracy

at $D_x \leq 0.1$

$\pm E/100$

at $D_x > 0.1$

$\pm E \times (1 + D_x)/100$

G accuracy

at $D_x \leq 0.1$

$\pm E/D_x (\%)$

at $D_x > 0.1$

$\pm E \times \sqrt{(1 + D_x^2)} / D_x (\%)$

Note: measurement accuracy is specified
under the following conditions:

Temperature: 23 °C ± 5 °C

Integration time: 1 PLC or 16 PLC

Parameters E_{POSC} Z_{OSC}

Oscillator level	E_{POSC} (%)	Z_{OSC} (m Ω)
125 mV < $V_{\text{OSC}} \leq 250$ mV	$0.03 \times (250 / V_{\text{OSC}} - 1)$	$5 \times (250 / V_{\text{OSC}} - 1)$
64 mV < $V_{\text{OSC}} \leq 125$ mV	$0.03 \times (125 / V_{\text{OSC}} - 1)$	$5 \times (125 / V_{\text{OSC}} - 1)$
32 mV < $V_{\text{OSC}} \leq 64$ mV	$0.03 \times (64 / V_{\text{OSC}} - 1)$	$5 \times (64 / V_{\text{OSC}} - 1)$
$V_{\text{OSC}} \leq 32$ mV	$0.03 \times (32 / V_{\text{OSC}} - 1)$	$5 \times (64 / V_{\text{OSC}} - 1)$

V_{OSC} is oscillator level in mV.

Parameters E_{PL} Y_{OL} Z_{SL}

Cable length	E_{PL} (%)	Y_{OL} (nS)	Z_{SL} (m Ω)
1.5 m	$0.02 + 3 \times f/100$	$750 \times f/100$	5.0
3 m	$0.02 + 5 \times f/100$	$1500 \times f/100$	5.0

f is frequency in MHz. If measurement cable is extended, open compensation, short compensation, and load compensation must be performed.

Parameters Y_{OSC} Y_0 E_p Z_s

Frequency	Y_{OSC} (nS)	Y_0 (nS)	E_p (%)	Z_s (m Ω)
1 kHz $\leq f \leq 200$ kHz	$1 \times (125 / V_{\text{OSC}} - 0.5)$	1.5	0.095	5.0
200 kHz < $f \leq 1$ MHz	$2 \times (125 / V_{\text{OSC}} - 0.5)$	3.0	0.095	5.0
1 MHz < $f \leq 2$ MHz	$2 \times (125 / V_{\text{OSC}} - 0.5)$	3.0	0.28	5.0
2 MHz < f	$20 \times (125 / V_{\text{OSC}} - 0.5)$	30.0	0.28	5.0

f is frequency in Hz.

V_{OSC} is oscillator level in mV.

Example of calculated C/G measurement accuracy

Frequency	Measured capacitance	C accuracy ¹	Measured conductance	G accuracy ¹
5 MHz	1 pF	± 0.61 %	≤ 3 μ S	± 192 nS
	10 pF	± 0.32 %	≤ 31 μ S	± 990 nS
	100 pF	± 0.29 %	≤ 314 μ S	± 9 μ S
	1 nF	± 0.32 %	≤ 3 mS	± 99 μ S
1 MHz	1 pF	± 0.26 %	≤ 628 nS	± 16 nS
	10 pF	± 0.11 %	≤ 6 μ S	± 71 nS
	100 pF	± 0.10 %	≤ 63 μ S	± 624 nS
	1 nF	± 0.10 %	≤ 628 μ S	± 7 μ S
100 kHz	10 pF	± 0.18 %	≤ 628 nS	± 11 nS
	100 pF	± 0.11 %	≤ 6 μ S	± 66 nS
	1 nF	± 0.10 %	≤ 63 μ S	± 619 nS
	10 nF	± 0.10 %	≤ 628 μ S	± 7 μ S
10 kHz	100 pF	± 0.18 %	≤ 628 nS	± 11 nS
	1 nF	± 0.11 %	≤ 6 μ S	± 66 nS
	10 nF	± 0.10 %	≤ 63 μ S	± 619 nS
	100 nF	± 0.10 %	≤ 628 μ S	± 7 μ S
1 kHz	100 pF	± 0.92 %	≤ 63 nS	± 6 nS
	1 nF	± 0.18 %	≤ 628 nS	± 11 nS
	10 nF	± 0.11 %	≤ 6 μ S	± 66 nS
	100 nF	± 0.10 %	≤ 63 μ S	± 619 nS

1. The capacitance and conductance measurement accuracy is specified under the following conditions:

$D_x = 0.1$

Integration time: 1 PLC

Test signal level: 30 mVrms

At four-terminal pair port of MFCMU

Atto-sense and switch unit (ASU) specifications

AUX path specification

Maximum voltage

- 100 V (AUX input to AUX common)
- 100 V (AUX input to circuit common)
- 42 V (AUX common to circuit common)

Maximum current

- 0.5 A (AUX input to force output)

ASU supplemental information

Band width (at -3 dB)

- 30 MHz (AUX port)

SMU CMU unify unit (SCUU) and guard switch unit (GSWU) specifications

The SCUU multiplexes the outputs from two SMUs (MPSMUs and/or HRSMUs) and the CMU. The SCUU outputs are two sets of Kelvin triaxial ports (Force and Sense). The SCUU also allows the SMUs to act as DC bias sources in conjunction with the CMU. Special cables are available to connect the SMUs and CMU with the SCUU, and an auto-detect feature automatically compensates for the cable length going to the SCUU.

The GSWU contains a relay that automatically opens for IV measurements and closes for CV measurements, forming a guard return path to improve CV measurement accuracy.

Supported SMU

- MPSMU and HRSMU

For SCUU

Inputs:

- Triaxial ports: Force1, Sense1, Force2, and Sense2
- BNC ports: for MFCMU
- Control port: for MFCMU

Outputs:

- Triaxial ports: Force1/CMUH, Sense1, Force2/CMUL, and Sense2

- Control port: for GSWU

- LEDs: SMU/CMU output status indicator

Docking mode:

- Direct and indirect mode

For GSWU

Input:

- Control port: for SCUU
- Mini pin plug ports: Guard1, Guard2

Output:

- LED: Connection status indicator

SCUU supplemental information

SMU path:

- Offset current: < 20 fA
- Offset voltage: < 100 μ V at 300 sec
- Closed channel residual resistance: < 200 m Ω
- Channel isolation resistance: > 10^{15} Ω

CMU path:

Test signal

Signal output level additional errors

- (CMU bias, open load):
 - ± 2 % (direct docking)
 - ± 7 % (indirect docking)

Signal output level additional errors

- (SMU bias, open load):
 - ± 5 % (direct docking, ≥ 10 kHz)
 - ± 10 % (indirect docking, ≥ 10 kHz)

Output impedance: 50 Ω , typical

Signal level monitor additional errors

(open load):

- ± 2 % (CMU bias), direct docking
- ± 5 % (SMU bias), direct docking
- ± 7 % (CMU bias), indirect docking
- ± 10 % (SMU bias), indirect docking

DC bias function

DC voltage bias (CMU bias):

- Range: 0 to ± 25 V

- Resolution: 1 mV

Additional errors (for CMU bias):

- ± 100 μ V (open load)

DC voltage bias (SMU bias):

- Range: 0 to ± 100 V

- Resolution: 5 mV

Additional errors (for SMU voltage output accuracy): ± 100 μ V (open load)

DC bias monitor additional errors (open load):

- ± 20 mV, direct docking
- ± 30 mV, indirect docking

Output impedance:

- 50 Ω , typical

DC output resistance: 50 Ω (CMU bias),

- 130 Ω (SMU bias)

Measurement accuracy

Impedance measurement error is given by adding the following additional error E_e to the MFCMU measurement error.

$$E_e = \pm(A + Z_s/|Z_x| + Y_0|Z_x|) \times 100 \text{ (\%)}$$

Z_x : Impedance measurement value (Ω)

A: 0.05 % (direct docking) or
0.1 % (indirect docking)

Z_s : 500 + 500 $\times$ f (m Ω)

Y_0 : 1 + 1000 $\times$ f/100 (nS)
(direct docking, x2 for indirect docking)

Note: f is frequency in MHz.

When the measurement terminals are extended by using the measurement cable, the measurement accuracy is applied to the data measured after performing the open/short/load correction at the DUT side cable end.

Note: The error is specified under the following conditions:

Temperature: 23 $^{\circ}$ C ± 5 $^{\circ}$ C

Integration time: 1 PLC or 16 PLC

HV-SPGU (high voltage semiconductor pulse generator unit) module specification

Specifications

Number of output channels:
2 channels per module

Modes: pulse, constant, and freerun

Standard pulse mode:

- Two level pulse
- Three level pulse per one channel
- Pulse period: 30 ns to 10 s

Delay range: 0 s to 9.99 s

Delay resolution: 2.5 ns (minimum)

Output count: 1 to 1,000,000

Voltage monitor minimum sampling
period: 5 μ s

Trigger output:

Level: TTL

Timing: Synchronized with pulse period

Trigger width:

Pulse period $\times$ 1/2 (pulse period $\leq$ 10 μ s)

Maximum 5 μ s (pulse period $>$ 10 μ s)

SPGU supplemental information

Pulse width jitter: 0.001 % +150 ps

Pulse period jitter: 0.001 % +150 ps

Maximum slew rate: 1000 V/ μ s (50 Ω load)

Noise: 10 mV_{rms} (at DC output)

Advanced feature:

Voltage monitor: The HV-SPGU has a voltage monitor function to measure the voltage at the DUT terminal.

Measurement accuracy (open load):
 $\pm(0.1\% \text{ of reading} + 25 \text{ mV})$

Measurement resolution: 50 μ V

Note: Specified at 1 PLC (20 ms = (5 μ s sample + 5 μ s interval) $\times$ 2000 samples.)

Voltage compensation: The HV-SPGU can measure the impedance of DUT and adjust the output voltage according to the DUT impedance.

Pulse/DC output voltage and accuracy

Output voltage (Vout)	50 Ω load	–20 V to +20 V
	Open load	–40 V to +40 V
Accuracy ¹	Open load	$\pm(0.5\% + 50 \text{ mV})$
Amplitude resolution	50 Ω load	0.2 mV ($\pm 10 \text{ V range}$)
		0.8 mV ($\pm 40 \text{ V range}$)
	Open load	0.4 mV ($\pm 10 \text{ V range}$) 1.6 mV ($\pm 40 \text{ V range}$)
Output connectors		SMA
Source impedance		50 Ω^2
Short circuit current		800 mA peak (400 mA average ³)
Overshoot/ pre-shoot/ringing ⁴	50 Ω load	$\pm(5\% + 20 \text{ mV})$
Output limit		Monitoring over current limit

1. At 1 μ s after completing transition.
2. Typical ($\pm 1\%$)
3. This value is specified under the following condition: [(Number of installed HV-SPGUs) $\times$ 0.2 A] + [DC current output by all modules (including HV-SPGUs)] $<$ 3.0 A
4. Following the specified condition with transition time.

Pulse range and pulse parameter¹

Frequency range		0.1 Hz to 33 MHz
Pulse period	Programmable range	30 ns to 10 s
	Resolution	10 ns
	Minimum	100 ns ³
	Accuracy	$\pm 1\%$ ($\pm 0.01\%$ ²)
Width	Programmable range	10 ns to (period – 10 ns)
	Resolution	2.5 ns (Tr and Tf $\leq$ 8 μ s) 10 ns (Tr or Tf $>$ 8 μ s)
	Minimum	50 ns (25 ns typical) ³
	Accuracy	$\pm(3\% + 2 \text{ ns})$
Transition time ⁵ (Tr and Tf)	Programmable range	8 ns to 400 ms
	Resolution	2 ns (Tr and Tf $\leq$ 8 μ s) 8 ns (Tr or Tf $>$ 8 μ s)
	Minimum (typical)	$< 15 \text{ ns}^3$
	Minimum	20 ns (Vamp $\leq$ 10 V) 30 ns (Vamp $\leq$ 20 V) 60 ns (Vamp $>$ 20 V)
	Accuracy	–5 % to 5 % + 10 ns (Vamp $\leq$ 10 V) –5 % to 5 % + 20 ns (Vamp $\leq$ 20 V)
Output relay switching time ⁴	Open/close	$< 100 \mu$ s

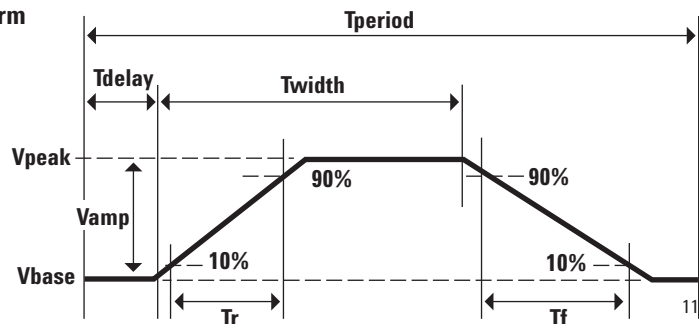
1. Unless otherwise stated, all specifications assume a 50 Ω termination.
2. Typical minimum. This is supplemental information.
3. This is specified at Vamp $\leq$ 10 V.
4. The time it takes the open state relay to open or close.
5. The time from 10 % to 90 % of Vamp which is the amplitude of output pulse.

ALWG (arbitrary linear waveform generator) function

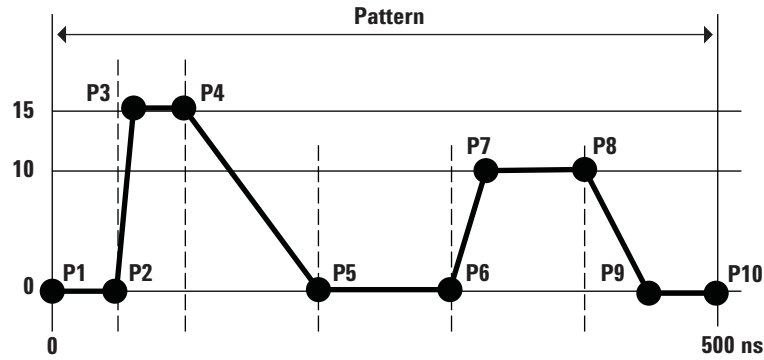
Arbitrary linear waveform generator (ALWG) mode:

- Output complex waveform per one channel of HV-SPGU
- Define multi-level pulse and multi-pulse waveform including open state pulse with ALWG GUI editor
- Sequential pulse waveform from user-defined pulse waveform
- 1024 points per one channel
- Programmable timing range:
10 ns to 671.088630 ms, 10 ns resolution

Pulse waveform

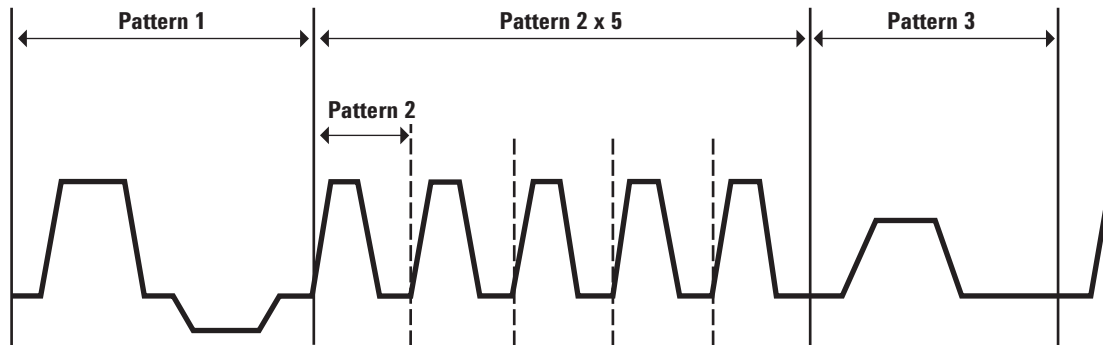


Example 1. ALWG setup table and pattern



Point	Time	Voltage
1	0	0.0 V
2	50 ns	0.0 V
3	70 ns	15.0 V
4	100 ns	15.0 V
5	200 ns	0.0 V
6	300 ns	0.0 V
7	320 ns	10.0 V
8	400 ns	10.0 V
9	450 ns	0.0 V
10	500 ns	0.0 V

Example 2. ALWG complex waveform



16440A SMU/pulse generator selector

The Agilent 16440A SMU/pulse generator selector switches either a SMU or PGU to the associated output port. You can expand to four channels by adding an additional 16440A. The PGU port on channel 1 provides a "PGU OPEN" function, which can disconnect the PGU by opening a semiconductor relay. The Agilent B1500A and 16445A are required to use the 16440A.

The following specifications data is specified at 23 °C ± 5 °C and 50% relative humidity.

- Channel configuration:
2 channels (CH 1 and CH 2).
Can add an additional 2 channels (CH 3 and CH 4) by adding another 16440A (selector expander).

	Input	Output
Channel 1 (CH 1)	2 (SMU and PGU ¹)	1
Channel 2 (CH 2)	2 (SMU and PGU)	1
Channel 3 (CH 3) ²	2 (SMU and PGU ¹)	1
Channel 4 (CH 4) ²	2 (SMU and PGU)	1

- PGU channels 1 & 3 have a built-in series semiconductor relay.
- Available when a second 16440A (selector expander) is installed.

- Voltage and current range

Input port	Maximum voltage	Maximum current
SMU	200 V	1.0 A
PGU	40 V	0.4 A ¹

- This is peak-to-peak current.

16445A SMU/PGU selector connection adaptor

The Agilent 16445A selector adapter is required to control and to supply DC power to the Agilent 16440A SMU/pulse generator selector.

Power requirement: 100 to 240 V, 50/60 Hz

Maximum volt-amps (VA): 20 VA

WGFMU (waveform generator/fast measurement unit) module specification

Overview

The WGFMU is a self-contained module offering the combination of arbitrary linear waveform generation (ALWG) with synchronized fast current or voltage (IV) measurement. The ALWG function allows you to generate not only DC, but also various types of AC waveforms. In addition to this versatile sourcing capability, the WGFMU can also perform measurement in synchronization with the applied waveform, which enables accurate high-speed IV characterization.

Specifications

Number of output channels:
2 channels per module

Modes: Fast IV, PG (pulse generator), DC, and SMU pass-through

RSU:

Output Connector: SMA
Source Impedance: 50 Ω (nominal) at DC in PG mode

SMU path: Maximum voltage ± 25 V,
Maximum current ± 100 mA

V monitor terminal:
Connector: BNC

Source Impedance: 50 Ω (nominal) at DC

The terminal outputs a buffered signal equal to 1/10 of Vout (into a 50 Ω load)

WGFMU to RSU cable length:

The WGFMU and RSU are connected by a special composite cable. The following configurations are available:

- 3 m
- 5 m
- 1.5 m
- 2.4 m + connector adapter + 0.6 m
- 4.4 m + connector adapter + 0.6 m

Note: The connector adapter is used when routing the cable through the prober's connector panel.

Measurement functions, voltage forcing, voltage measurement, and current measurement

Mode	Function	V force ranges	V measure ranges	I measure ranges
Fast IV	V force/I measure, V force/V measure	-3 V to +3 V -5 V to +5 V -10 V to 0 V 0 V to +10 V	-5 V to +5 V -10 V to +10 V	1 μ A, 10 μ A, 100 μ A, 1 mA, 10 mA.
PG	V force/V measure	-3 V to +3 V -5 V to +5 V	-5 V to +5 V	—
DC	V force/I measure, V force/V measure	-3 V to +3 V -5 V to +5 V -10 V to 0 V 0 V to +10 V	-5 V to +5 V -10 V to 10 V	1 μ A, 10 μ A, 100 μ A, 1 mA, 10 mA
SMU pass-through	Measurement using SMU	Max ± 25 V	—	Max ± 100 mA

Voltage force accuracy, resolution, and timing

V force (Fast IV mode)	-5 V to 5 V, -10 V to 0 V, 0 V to 10 V
V force (PG mode)	-5 V to 5 V (open load) -2.5 V to 2.5 V (50 Ω load)
Accuracy	$\pm 0.1\%$ of setting $\pm 0.1\%$ of range ¹
Resolution ²	96 μ V (-3 to 3V) 160 μ V (all ranges except for -3 V to 3 V)
Overshoot/undershoot	$\pm(5\%+20$ mV) ³
Noise	Maximum 0.1 mV _{rms} ⁴
Rise time T _{rise} (10 to 90%)/ Fall time T _{fall} (90 to 10%)	Accuracy: -5% to (+5% +10 ns) of setting ⁵ Minimum: 24 ns, PG mode and 50 Ω load
Pulse period	Timing Accuracy: $\pm 1\%$ of setting ⁶ Minimum: 100 ns, PG mode and 50 Ω load
Pulse width	Accuracy: $\pm(3\%+2$ ns) ⁷ Minimum: 50 ns, PG mode and 50 Ω load

Voltage measurement accuracy, resolution, and noise

Accuracy	$\pm(0.1\%$ of reading $\pm 0.1\%$ of range) ⁸
Resolution ⁹	680 μ V (-5 V to +5 V range) 1.4 mV (-10 V to +10 V range)
Noise ¹⁰	Maximum 4 mV _{rms} (-5 V to +5 V range)

1. Independent of the range or the mode. DC constant voltage output. Load impedance must be ≥ 1 M Ω (1 μ A range) or ≥ 200 k Ω (all other current ranges) for Fast IV mode, or ≥ 1 M Ω for PG mode.
2. Can vary at most 5% based on the result of calibration.
3. PG mode, 50 Ω load, T_{rise} and T_{fall} >16 ns with the 1.5 m cable, >32 ns with 3 m cable, or >56 ns with 5 m cable
4. Theoretical value for observed time 100 ns to 1 ms, supplemental information
5. PG mode, 50 Ω load, T_{rise} and T_{fall} ≥ 24 ns
6. PG mode, 50 Ω load, pulse period ≥ 100 ns
7. PG mode, 50 Ω load, pulse width ≥ 50 ns
8. Independent of the range or the mode. DC constant voltage output. Applicable condition: 10,000 averaging samples for 10 μ A range and above; 100,000 averaging samples for the 1 μ A range.
9. Display resolution. Can vary at most 5% based on the result of calibration.
10. 0 V output, open load, no averaging. Maximum 1.5 mV_{rms} as supplemental information.

Current measurement accuracy and resolution

Accuracy	$\pm(0.1\% \text{ of reading } \pm 0.2\% \text{ of range})^1$
Resolution ²	0.014% of range
Noise (Effective resolution)	Maximum 0.2% of range ³

1. Independent of the range or the mode. DC constant voltage output. Applicable condition: 10,000 averaging samples for 10 μA range and above; 100,000 averaging samples for the 1 μA range.
2. Display resolution. Can vary at most 5% based on the result of calibration
3. Effective value at 0 V output, open load, and no averaging. Supplemental information

ALWG function

Maximum number of vectors	2048
Maximum number of sequences	512
Maximum number of loop counts	1 to 10^{12}
Length of a vector	10 ns to 10,000 s with 10 ns resolution
Sampling rate	5 ns, or 10 ns to 1 s with 10 ns resolution
Averaging time	10 ns to 20 ms with 10 ns resolution
Hardware memory	About 4 M data points/channel (typical)

Trigger output

Level: TTL

Trigger width: 10 ns

Generated synchronously with ALWG waveform.

Supplemental Information

RSU SMU path:

Leak current: < 100 pA

Residual resistance: <300 m Ω

Jitter: <1 ns

Skew between channels: <3 ns, under no electrostatic discharge condition.

Trigger output skew: <3 ns

Current range change time: <150 μs *

* The time until the measured current settles within $\pm 0.3\%$ of the final result value after the range change.

Minimum timing parameters for current measurement¹

Voltage applied to DUT		10 V					
Current applied to DUT		100 nA	1 μA	10 μA	100 μA	1 mA	10 mA
Applied voltage condition	Recommended minimum pulse width ²	47 μs	38.7 μs	6.8 μs	950 ns	240 ns	145 ns
	Measurement Range	1 μA	1 μA	10 μA	100 μA	1 mA	10 mA
	Recommended minimum measurement window	10 μs	1.64 μs	1 μs	130 ns	40 ns	20 ns
	Settling time ³	37 μs	37 μs	5.8 μs	820 ns	200 ns	125 ns
	Noise (rms) ⁴	160 pA	425 pA	2.5 nA	47 nA	280 nA	1.9 μA

1. Measurement conditions: The DUT is a resistive load chosen to adjust the flowing current to the specified current in the table above. The capacitance of the cable between the RSU and the DUT is 20 pF. Voltage is applied to the DUT by a channel of WGFMU/RSU in Fast IV mode and in the 10 mA range, and current measurement is performed by another channel at 0 V in Fast IV mode.
2. Recommended minimum pulse width = settling time + recommended minimum measurement window.
3. The time until the measured value settles to within $\pm 0.6\%$ of the final result value after the output voltage is changed from the initial value (0 V). Minimum rise/fall time of 70 ns is recommended for minimizing overshoot.
4. RMS noise measured over the recommended minimum measurement window.

Minimum timing parameters for voltage measurement¹

Voltage applied to DUT		5V	10 V
Applied voltage condition	Recommended minimum pulse width ²	105 ns	130 ns
	Measurement Range	5 V	10 V
Voltage measurement condition	Recommended minimum measurement window	20 ns	20 ns
	Settling time ³	85 ns	110 ns
	Noise (rms) ⁴	1.4 mV	1.4 mV

1. Measurement conditions: The DUT is a resistive load between 1 k Ω and 10 M Ω . The capacitance of the cable between the RSU and the DUT is 20 pF. Voltage is applied to the DUT by a channel of WGFMU/RSU, and voltage measurement is performed by the same channel. (PG mode for 5 V, Fast IV mode for 10 V)
2. Recommended minimum pulse width = settling time + recommended minimum measurement window.
3. The time until the measured value settles to within $\pm 0.6\%$ of the final result value after the output voltage is changed from the initial value (0 V). Minimum rise/fall time of 70 ns for 10 V, or 30ns for 5 V is recommended for minimizing overshoot.
4. RMS noise measured over the recommended minimum measurement window.

Software

Instrument library for WGFMU control

Operating system:

Microsoft Windows XP Professional SP3
or later, Windows Vista Business SP2
or later (32bit only), and Windows 7
Professional SP1 or later (32 bit and 64bit)

Supported language: English (US)

NBTI and general-purpose EasyEXPERT
Application Tests

Sample programs (NBTI and general-
purpose measurement using WGFMU and
RTS data analysis)

WGFMU supported probe vendors

Cascade Microtech

Suss MicroTec

Vector Semicon

Note: The maximum number of installable RSUs for a given probe depends upon the available space. Please contact your local sales representative for details on connecting and mounting the WGFMU and RSU.

Agilent EasyEXPERT software

Functions

Operation mode:

- Application test mode, Classic test mode, Quick test mode

Key features

- Categorized and predefined application test library
- GUI-based application test editor
- Save/Recall "My Favorite Setups"
- Define/customize application library
- Execute measurement (Single/Repeat/Append)
- Quick test execution
- Direct control (GPIB FLEX)
- Save/Recall measurement data and settings
- Test result data management
- Import/Export device definition, measurement settings, my favorite setup, measurement data, and application library
- Graph plot display/analysis/printing
- Switching matrix control
- Workspace management
- Self-test, self-calibration, diagnostics

Application library

Sample application tests are supplied for the following categories; they are subject to change without notice.

Structure, CMOS, Bipolar (BJT), Memory, Mixed Signal Device, TFT, Discrete, Reliability, Power Device, Nanotechnology, Utility

Measurement mode details

The Agilent B1500A supports the following measurement modes:

- Staircase sweep
- Multi-channel sweep¹
- Pulsed sweep
- Staircase sweep with pulsed bias
- IV sampling
- High speed IV sampling
- CV sweep
- C-t sampling
- C-f sweep
- List sweep
- Linear search²
- Binary search²

1. EasyEXPERT does not support VAR1' in multi-channel sweep mode.
2. They are supported by FLEX command only.

Each SMU can be set to VAR1 (primary sweep), VAR2 (secondary sweep), VAR1' (synchronous sweep), or CONST (constant voltage/current source).

VAR1

Primary sweep controls the staircase (DC or pulsed) voltage or current sweep. Maximum number of VAR1 steps: $N_1 = 1001$

VAR2

Subordinate linear staircase or linear pulsed sweep. After primary sweep is completed, the VAR2 unit output voltage or current is changed.

Maximum number of VAR2 steps: $N_2 = 1001$ ($1 \leq N_1 \times N_2 \leq 128128$)

VAR1'

Staircase or pulse sweep synchronized with the VAR1 sweep. Sweep is made with a user specified ratio and offset value. VAR1' output is calculated as $VAR1' = a \times VAR1 + b$, where "a" is the user specified ratio and "b" is the user specified offset value.

CONST

A source unit can be set as a constant voltage or current source depending on the unit.

Staircase sweep measurement mode

Forces swept voltage or current, and measures DC voltage or current. One channel can sweep current or voltage while up to ten channels can measure current or voltage. A second channel can be synchronized with the primary sweep channel as an additional voltage or current sweep source.

Number of steps: 1 to 1001

Sweep mode: Linear or logarithmic (log)

Sweep direction: Single or double sweep

Hold time: 0 to 655.35 s, 10 ms resolution

Delay time: 0 to 65.5350 s, 100 μ s resolution

Pulsed sweep measurement mode:

Forces pulsed swept voltage or current, and measures DC voltage or current. A second channel can be programmed to output a staircase sweep voltage or current synchronized with the pulsed sweep output.

Staircase sweep with pulsed bias measurement mode

Forces swept voltage or current, and measures DC voltage or current. A second channel can be programmed to output

a pulsed bias voltage or current. A third channel can be synchronized with the primary sweep channel as an additional voltage or current sweep source.

Sampling (time domain) measurement mode

Displays the time sampled voltage/current data (by SMU) versus time.

Sampling channels: Up to 10

Sampling mode: Linear, logarithmic (log)

Sampling points:

For linear sampling:

1 to 100,001/(number of channels)

For log sampling:

1 to 1+ (number of data for 11 decades)

Sampling interval range:

100 μ s +20 μ s x (num. of channels – 1)
to 2 ms, 10 μ s resolution
2 ms to 65.535 s, 1 ms resolution

Hold time, bias hold time:

-90 ms to -100 μ s, 100 μ s resolution
0 to 655.35 s, 10 ms resolution

Measurement time resolution: 100 μ s

Standby mode

SMUs in "Standby" remain programmed to their specified output value even as other units are reset for the next measurement.

Current offset cancel

This function subtracts the offset current from the current measurement raw data, and returns the result as the measurement data. This function is used to compensate the error factor (offset current) caused by the measurement path such as the measurement cables, manipulators, or probe card.

Time stamp

The B1500A supports a time stamp function utilizing an internal quartz clock. Resolution: 100 μ s

Other measurement characteristics

Measurement control:

Single, repeat, append, and stop

SMU setting capabilities:

Limited auto ranging, voltage/current compliance, power compliance, automatic sweep abort functions, self-test, and self-calibration

Arithmetic and analysis functions

User functions

Up to 20 user-defined functions can be defined using arithmetic expressions.

Measured data and pre-defined variables can be used in the computation. The results can be displayed on the LCD.

Arithmetic operators

+, -, *, /, ^, abs (absolute value), at (arc tangent), avg (averaging), cond (conditional evaluation), delta, diff (differential), exp (exponent), integ (integration), lgt (logarithm, base 10), log (logarithm, base e), mavg (moving average), max, min, sqrt, trigonometric function, inverse trigonometric function, and so on.

Physical constants

Keyboard constants are stored in memory as follows:

q: Electron charge, 1.602177E-19 C

k: Boltzman's constant, 1.380658E-23

ε (e): Dielectric constant of vacuum, 8.854188E-12

Engineering units

The following unit symbols are also available on the keyboard:

a (10^{-18}), f (10^{-15}), p (10^{-12}), n (10^{-9}),
u or μ (10^{-6}), m (10^{-3}), k (10^3), M (10^6),
G (10^9), T (10^{12}), P (10^{15})

Analysis capabilities

Overlay graph comparison

Graphical plots can be stored and overlaid.

Scale

Auto scale and zoom

Marker

Marker to min/max, interpolation, direct marker, and marker skip

Cursor

Direct cursor

Line

Two lines, normal mode, grad mode, tangent mode, and regression mode

Automatic analysis function

On a graphics plot, the markers and lines can be automatically located using the auto analysis setup. Parameters can be automatically determined using automatic analysis, user function, and read out functions.

Data variable display

Up to 20 user-defined parameters can be displayed on the graphics screen.

Analysis functions

Up to 20 user-defined analysis functions can be defined using arithmetic expressions.

Measured data, pre-defined variables, and read out functions can be used in the computation. The results can be displayed on the LCD.

Read out functions

The read out functions are built-in functions for reading various values related to the marker, cursor, or line.

Graph plot

Display mode

Data display window can be printed. Only X-Y graph can be printed.

Graph plot file

Graph plot can be stored as image data to clip board or mass storage device.

File type: bmp, gif, png, emf

Output

Display mode

X-Y graph, list, and parameter

X-Y graph display

X-axis and up to eight Y-axes, linear and log scale, real time graph plotting

List display

Measurement data and calculated user function data are listed in conjunction with VAR1 step number or time domain sampling step number. Up to 20 data sets can be displayed.

Other functions

Import/export files

File type:

Agilent EasyEXPERT format, XML-SS format, CSV format

Data storage

Hard disk drive, DVD-ROM/CD-R/CD-RW drive

Interfaces

GPIO, interlock, USB (USB 2.0, front 2, rear 2), LAN (100BASE-TX/10BASE-T), trigger in/out, digital I/O

Trigger I/O

Only available using GPIO FLEX commands.

Trigger in/out synchronization pulses before and after setting and measuring DC voltage and current. Arbitrary trigger events can be masked or activated independently.

Supported external instruments

EasyEXPERT Standard edition:

- Supported by switching matrix GUI: B2200A/B2201A
- Supported by application tests: E5250A (E5252A cards), 4284A/E4980A, 81110A, 3458A

EasyEXPERT Plus edition:

- All external instruments supported by EasyEXPERT Standard edition
- Also supported by switching matrix GUI: E5250A (E5252A cards)

Furnished software

- Prober control execution files
- Desktop EasyEXPERT software with license-to-use for Standard edition
- 4155/56 setup file converter tool (Supported operating systems: Microsoft® Windows® XP Professional (SP3 or later), Windows Vista Business (SP2 or later), and Windows 7 Professional (SP1 or later); Supported language: English (US))
- A VXIplug&play driver for the B1500A (Supported operating systems: Microsoft® Windows® XP Professional (SP3 or later), Windows Vista Business (SP2 or later), and Windows 7 Professional (SP1 or later); Supported language: English (US))

Agilent Desktop EasyEXPERT software

Desktop EasyEXPERT is the same software that is built-in to the PC-based Agilent B1500A Semiconductor Device Analyzer, except that it runs on a stand-alone PC. Just like standard EasyEXPERT, Desktop EasyEXPERT supports all aspects of parametric test, from basic manual measurements to test automation across a wafer in conjunction with a semi-automatic wafer prober.

Features and benefits

Large application test library

Desktop EasyEXPERT comes with over 300 application tests conveniently organized by device type, application, and technology. Many of these application tests will run on the 4155/4156 without modification, and you can easily edit and customize the furnished application tests to fit your specific needs.

Offline capability

Desktop EasyEXPERT can be run in either online or offline mode. In the offline mode you can perform tasks such as analyzing data and creating new application tests. This frees up your existing analyzer from being needed for development work and enables you to use it for its primary purpose: making measurements.

GUI-based classic test mode

Desktop EasyEXPERT offers a classic test mode that maintains the look, feel, and terminology of the 4155/4156 user interface. In addition, it improves the 4155/4156 user interface by taking full advantage of Microsoft Windows GUI features.

Easy test sequencing

A GUI-based Quick Test mode enables you to perform test sequencing without programming. You can select, copy, rearrange and cut-and-paste any application tests with a few simple mouse clicks. Once you have selected and arranged your tests, simply click on the measurement button to begin running an automated test sequence.

Prober control

All popular semiautomatic wafer probers are supported by Desktop EasyEXPERT. You can define wafer, die, and module information for probing across an entire wafer. You can also combine wafer prober control with either Quick Test mode or an application test based test sequence to perform multiple testing on various devices across the wafer.

Automatic data export

The Desktop EasyEXPERT has the ability to automatically export measurement data in real time, in a variety of formats. You can save data to any drive connected to the PC. If you wish, you can export data to a network drive and view test results on your desktop PC as your instruments are performing the testing in your lab.

System requirements

The following are the minimum requirement for executing Desktop EasyEXPERT.

Operating system and service pack	Microsoft Windows XP Professional SP3 or later	Microsoft Windows Vista Business SP2 or later	Microsoft Windows 7 Professional SP1 or later
Supported language	English (US)		
Processor	Intel Celeron 2 GHz	Vista certified PC	Windows 7 certified PC
Memory	512 MB DDR266	1 GB memory	2 GB memory
Display	XGA 1024 x 768 (SXGA 1280 x 1024 recommended)		
HDD	1GB free space on the C drive, 10GB (30GB recommended) free space on a drive for test setup / result data storage.		
.NET Framework	Microsoft .NET Framework 2.0 SP2		
IO Libraries (for online mode)	Agilent IO Libraries Suite 16.1 or later		

Supported GPIB I/F (for online mode)

	B1500A	4155B/C 4156B/C
Agilent 82350B	0	0
Agilent 82357A	X	0
Agilent 82357B	X	0

0 = Supported
X = Not supported

Supported instruments

- B1500A
- 4155B, 4156B, 4155C, and 4156C
- Supported 4155/4156 firmware:
HOSTC: 03.08 or later
SMUC: 04.08 or later

Supported external instruments

Desktop EasyEXPERT Standard edition:

- Supported by switching matrix GUI:
B2200A/B2201A
- Supported by application tests: E5250A (E5252A cards), 4284A/E4980A, 81110A, 3458A

Desktop EasyEXPERT Plus edition:

- All external instruments supported by Desktop EasyEXPERT Standard edition
- Also supported by switching matrix GUI: E5250A (E5252A cards)

Supported 4155/4156 functionality

Desktop EasyEXPERT Standard edition:

- I/V Sweep
- B2200A and B2201A switching matrix GUI control

Desktop EasyEXPERT Plus edition:

The following functions are additionally supported.

- I/V-t sampling (except thinned out mode)
- VSU/VMU (except differential voltage measurement using VMU)
- PGU (41501B)
- E5250A/E5252A switching matrix GUI control

Setup converter tool

In addition to Desktop EasyEXPERT, Agilent supplies a free setup converter tool that runs on any Windows-based PC. This tool can convert 4155 and 4156 measurement setup files (file extensions MES or DAT) into equivalent Desktop EasyEXPERT classic test mode setup files.

General specifications

Temperature range

Operating: +5 °C to +40 °C
Storage: -20 °C to +60 °C

Humidity range

Operating: 20 % to 70 % RH, non-condensing
Storage: 10 % to 90 % RH, non-condensing

Altitude

Operating: 0 m to 2,000 m (6,561 ft)
Storage: 0 m to 4,600 m (15,092 ft)

Power requirement

AC voltage: 90 V to 264 V
Line frequency: 47 Hz to 63 Hz

Maximum volt-amps (VA)

B1500A: 900 VA

Regulatory compliance

EMC:

IEC61326-1:+A1/EN61326-1:+A1
AS/NZS 2064.1

Safety:

CSA C22.2 No.1010.1-1992
IEC61010-1:+A2/EN61010-1:+A2
UL3111-1:1994

Certification

CE, CSA, NRTL/C, C-Tick

Dimensions

B1500A:

420 mm W x 330 mm H x 575 mm D

N1301A-100 SMU CMU unify unit:

148 mm W x 75 mm H x 70 mm D

N1301A-200 guard switch unit:

33.2 mm W x 41.5 mm H x 32.8 mm D

E5288A Atto-sense and switch unit:

132 mm W x 88.5 mm H x 50 mm D

B1531A RSU:

45.2 mm W x 70 mm H x 82 mm D

16440A SMU/PGU selector:

250 mm W x 50 mm H x 275 mm D

16445A Selector adaptor:

250 mm W x 50 mm H x 260 mm D

Weight

B1500A (empty): 20 kg

B1510A: 2.0 kg

B1511A: 1.0 kg

B1517A: 1.2 kg

B1520A: 1.5 kg

B1525A: 1.3 kg

B1530A: 1.3 kg

B1531A: 0.13 kg

E5288A: 0.5 kg

N1301A-100: 0.8 kg

N1301A-200: 0.1 kg

16440A: 1.1 kg

16445A: 1.0 kg

Furnished accessories

Power cable

Manual CD-ROM

Desktop EasyEXPERT CD-ROM

Software CD-ROM (including VXiplug&play driver and utility tools)

License-to-use for Desktop EasyEXPERT standard edition

Order information

Mainframe

B1500A	Semiconductor device analyzer mainframe
	The following accessories are included
16444A-001	Keyboard
16444A-002	USB mouse
16444A-003	Stylus pen
16493J-001/002	Interlock cable 1.5m or 3.0m*
16493L-001/002	GNDU cable 1.5m or 3.0m *
16494A-001/002	Tri-axial cable 1.5m or 3.0m *
N1254A-100	GNDU to Kelvin adaptor
CD-ROMs	Manuals, Others
	*Select B1500A-015 or B1500A-030 to specify cable length
B1500A-015	1.5m cable (Cable length is set to 1.5m for standard and add-on packages)
B1500A-030	3.0m cable (Cable length is set to 3.0m for standard and add-on packages)
B1500A-050	50 Hz line frequency
B1500A-060	60 Hz line frequency
B1500A-A6J	ANSI Z540 compliant calibration
B1500A-UK6	Commercial calibration certificate with test data
B1500A-ABA	English paper document
B1500A-ABJ	Japanese paper document

Standard packages

B1500A-A00	Empty Package for Custom Solution
B1500A-A01	Standard Package (MPSMU 4ea. & Cables)
B1500A-A02	High Resolution Package (HRSMU 2ea, MPSMU 2ea. & Cables)
B1500A-A03	High Power Package (HPSMU 2ea, MPSMU 2ea & Cables)
B1500A-A04	Basic Flash Memory Cell Package (MPSMU 2ea, HRSMU 2ea, SPGU, Accessories)

Add-on packages

B1500A-A10	HPSMU Add-on (HPSMU 1ea. & Cables)
B1500A-A11	MPSMU Add-on (MPSMU 1ea. & Cables)
B1500A-A17	HRSMU Add-on (HRSMU 1ea. & Cables)
B1500A-A20	MFCMU Add-on (MFCMU, Cable)
B1500A-A25	HVSPGU Add-on (HVSPGU 1ea. & Cables)
B1500A-A28	ASU (Atto Sense Unit) Add-on (ASU 1ea. & Cables)
B1500A-A30	WGFMU Add-on (WGFMU 1ea. RSU 2ea. & Cables)
B1500A-A31	WGFMU Add-on with Connector Adapter (WGFMU 1ea, RSU 2ea, Cables & Connector Adapter)
B1500A-A3P	WGFMU probe cable kit (8 probe cables. WGFMU is not included.)
B1500A-A5F	Test fixture for packaged device measurement (16442B 1ea)

Other accessories

N1301A	CMU Accessories for B1500
N1301A-100	SMU CMU unify unit (SCUU)
N1301A-102	SMU CMU unify unit cable (3m)
N1301A-110	SMU CMU unify unit magnetic stand
N1301A-200	Guard switch unit (GSWU)
N1301A-201	Guard switch unit cable (1 m)
N1301A-202	Guard switch unit cable (3 m)
B1540A	EasyEXPERT Software and Measurement Libraries
B1540A-001	Agilent EasyEXPERT with license-to-use for standard version
B1540A-002	License-to-use for Agilent EasyEXPERT Plus
B1541A	Desktop EasyEXPERT software and measurement libraries
B1541A-001	Agilent Desktop EasyEXPERT with license-to-use for standard version
B1541A-002	License-to-use for Agilent Desktop EasyEXPERT Plus
B1542A	Pulse IV Package for B1500 / EasyEXPERT

Package Option Contents ¹

Standard packages

B1500A-A01 Standard package

Item	Description	Qty
B1511A	MPSMU (Medium Power SMU)	4
16494A-001/002	Triaxial cable 1.5m or 3.0m	8

B1500A-A02 High resolution package

Item	Description	Qty
B1517A	HRSMU (High Resolution SMU)	4
16494A-001/002	Triaxial cable 1.5m or 3.0m	8

B1500A-A03 High power package

Item	Description	Qty
B1511A	MPSMU (Medium Power SMU)	2
B1510A	HPSMU (High Power SMU)	2
16494A-001/002	Triaxial cable 1.5m or 3.0m	8

B1500A-A04 Basic flash memory cell package

Item	Description	Qty
B1511A	MPSMU (Medium Power SMU)	2
B1517A	HRSMU (High Resolution SMU)	2
B1525A	HVSPGU (Pulse Generator Unit)	1
16493P-001 / 002	SPGU CABLE (SMA-TO-COAXIAL) 1.5m or 3.0m	2
16440A	SMU/PGU Pulse Selector	1
16440A-003	Control Cable 40cm (2nd Selector)	1
16445A	Selector Connection Unit	1
16445A-001	Control Cable For B1500A To 16440A 1.5m	1
16494A-001	Tri-axial cable 1.5m	2
16494A-001/002	Triaxial cable 1.5m or 3.0m	8

Add-on packages

B1500A-A10 HPSMU Add-on package

Item	Description	Qty
B1510A	HPSMU (High Power SMU)	1
16494A-001/002	Triaxial cable 1.5m or 3.0m	2

B1500A-A11 MPSMU Add-on package

Item	Description	Qty
B1511A	MPSMU (Medium Power SMU)	1
16494A-001/002	Triaxial cable 1.5m or 3.0m	2

B1500A-A17 HRSMU Add-on package

Item	Description	Qty
B1517A	HRSMU (High Resolution SMU)	1
16494A-001/002	Triaxial cable 1.5m or 3.0m	2

B1500A-A20 MFCMU Add-on package

Item	Description	Qty
B1520A	MFCMU	1
N1300A-001/002	CMU cable for B1500A 1.5m or 3.0m	1

B1500A-A25 HVSPGU Add-on package

Item	Description	Qty
B1525A	HVSPGU	1
16493P-001/002	SPGU cable (SMA to Coaxial) 1.5m or 3.0m	2

B1500A-A28 ASU Add-on package

Item	Description	Qty
E5288A	ASU	1
E5288A-001/002	Triaxial and Dsub cable for ASU 1.5m or 3.0m	1

B1500A-A30 WGFMU Add-on package ²

Item	Description	Qty
B1530A	One WGFMU and two RSUs	1
B1530A-005/002	Two WGFMU cables (1.5m or 3.0m) to connect between WGFMU and RSU	1

B1500A-A31 WGFMU Add-on package with connection adapter ²

Item	Description	Qty
B1530A	One WGFMU and two RSUs	1
B1530A-001	Two set of WGFMU cables (0.6m + 2.4m)	1
16493R-801	WGFMU connector adapter	2

B1500A-A3P WGFMU Probe cable kit

Item	Description	Qty
16493R-101	SSMC-SSMC cable (50mm) for current return path	2
16493R-102	SSMC-SSMC cable (70mm) for current return path	2
16493R-202	SMA-SSMC cable (200mm) between RSU and DC probe	2
16493R-302	SMA-SMA cable (200mm) between RSU and RF probe	2

B1500A-A5F Test fixture for packaged device measurement

Item	Description	Qty
16442B	Test fixture	1
	Test fixture adapter	1
	Universal socket module	2
	28 pin DIP socket module	1
	Blank Teflon board	1
	Cables used in test fixture adapter	39
	Carrying case	1

1. Cable length is set by B1500A-015 or B1500A-030 option

2. Order 16493R-802 if magnet stand is necessary for RSU



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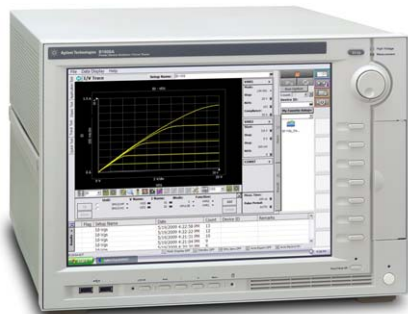
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